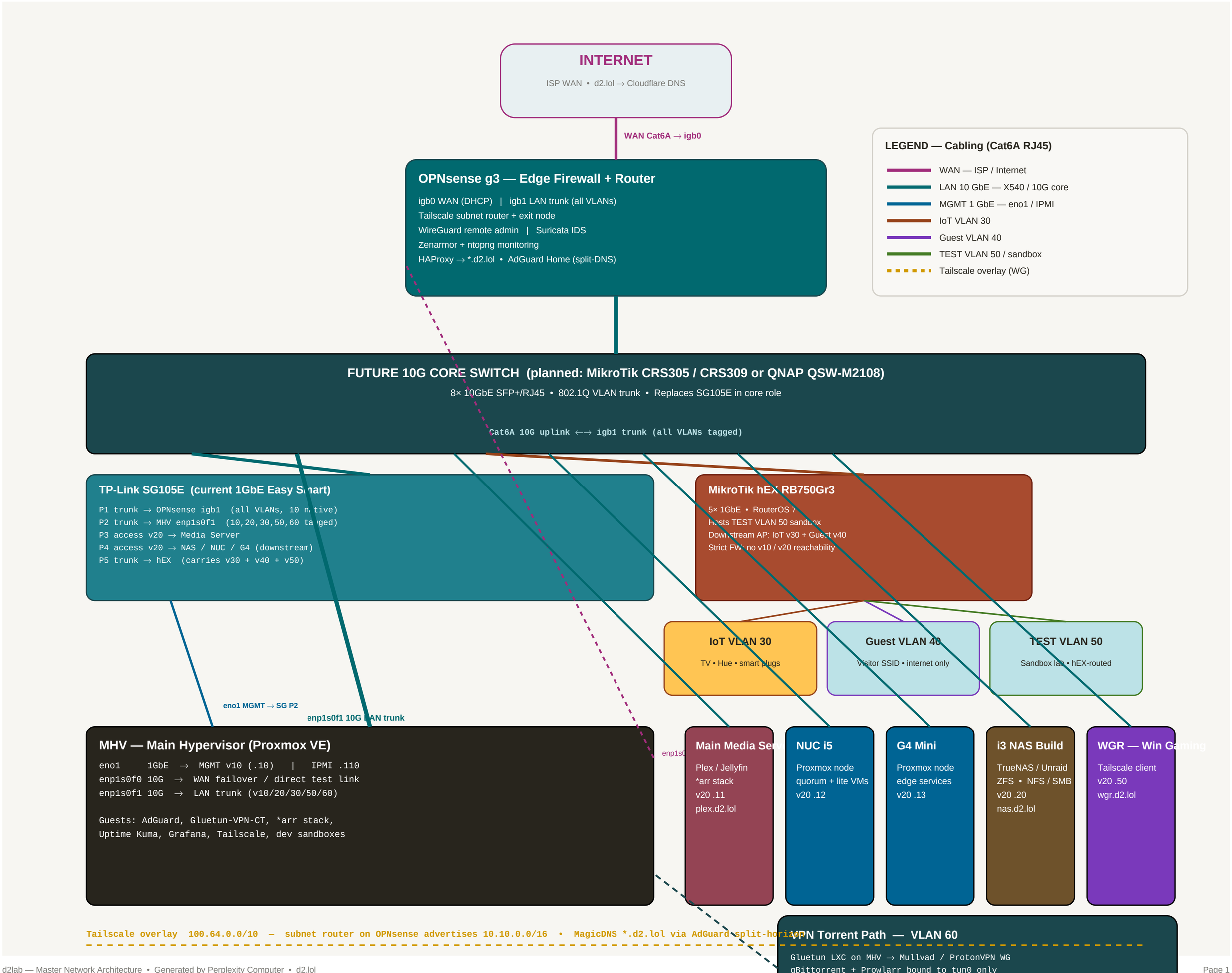
d2lab — Master Homelab Network Architecture

Domain d2.lol • Hypervisor MHV (Proxmox VE) • Cabling Cat6A RJ45 • Tailscale tailnet enabled • Future 10G core



VLAN & IP Plan

VLAN	Name	Subnet	Gateway	Purpose
10	MGMT	10.10.10.0/24	.1	Proxmox eno1, IPMI, switch admin, OPNsense mgmt
20	LAN	10.10.20.0/24	.1	Servers, NAS, gaming rig, *arr stack consumers
30	IoT	10.10.30.0/24	.1	TV, Hue Bridge, smart plugs — no LAN reach
40	GUEST	10.10.40.0/24	.1	Visitor SSID — internet only, fully isolated
50	TEST	10.10.50.0/24	.1	Sandbox / break-things lab routed via hEX
60	VPN	10.10.60.0/24	.1	Gluetun torrent egress — policy-routed, kill-switched
—	TS	100.64.0.0/10	—	Tailscale overlay — MagicDNS + split-horizon d2.lol

Static Host Assignments

Host	VLAN(s)	IP(s)	DNS (d2.lol)	Notes
OPNsense g3	10/20/30/40/50/60 trunk	all .1 + ISP WAN	fw.d2.lol	Edge FW, Tailscale exit, AdGuard
TP-Link SG105E	10 mgmt	10.10.10.2	sw1.d2.lol	Easy-Smart, 802.1Q trunks
MikroTik hEX RB750Gr3	10 mgmt + 30/40/50	10.10.10.3	hex.d2.lol	Test-net router, IoT/Guest gw
Future 10G core sw	10 mgmt	10.10.10.4	sw10g.d2.lol	CRS305/CRS309 or QSW-M2108
MHV (Proxmox)	10 + 20 + IPMI	eno1 .10 • enp1s0f1 .10 • IPMI .110	mhv.d2.lol	X540 dual 10G, eno1 mgmt
Main Media Server	20 (IPMI on 10)	10.10.20.11 • IPMI .111	plex.d2.lol	Plex/Jellyfin + *arr
NUC i5	20	10.10.20.12	nuc.d2.lol	Proxmox node, quorum
G4 Mini	20	10.10.20.13	g4.d2.lol	Proxmox node, edge svcs
i3 NAS Build	20	10.10.20.20	nas.d2.lol	TrueNAS/Unraid, ZFS
WGR (Windows)	20	10.10.20.50	wgr.d2.lol	Tailscale client
TV / Hue Bridge	30	10.10.30.30 / .31	—	IoT — no v10/v20 reach
Visitor SSID pool	40	10.10.40.100–.200 DHCP	—	Captive portal optional
Gluetun-VPN-CT	60	10.10.60.60	vpn-ct.d2.lol	LXC on MHV, kill-switched

Cabling Layout & Port Map

All physical links are **Cat6A RJ45**. Color encodes role; runs marked 10G use the future 10G core switch (or X540 direct between MHV and OPNsense for the WAN-failover/test path).

MHV (Proxmox) Interfaces

Interface	Speed	Role	Cable	Other End	VLANs
eno1	1 GbE	Management	Cat6A	SG105E P2	v10 (untagged)
enp1s0f0	10G	WAN failover / direct	Cat6A	OPNsense (igb0 alt) — direct	WAN
enp1s0f1	10G	LAN trunk	Cat6A	Future 10G core (uplink 1)	v10/20/30/50/60 tagged
IPMI/BMC	100M	Out-of-band mgmt	Cat6A	SG105E P2 (shared mgmt)	v10

SG105E Port Map (5 ports, 802.1Q)

Port	Mode	VLANs	Connected To
P1	Trunk	10 (native), 20/30/40/50/60 tagged	OPNsense igb1
P2	Trunk	10 (native), 20/30/50/60 tagged	MHV enp1s0f1 (and shared eno1 mgmt)
P3	Access	v20	Main Media Server
P4	Access	v20	Downstream NUC / G4 / NAS
P5	Trunk	30/40/50 tagged	MikroTik hEX RB750Gr3

hEX RB750Gr3 Port Map

Port	Role	VLAN(s)	Connected To
ether1	Trunk uplink	30/40/50	SG105E P5
ether2	Access — TEST lab	v50	Lab patch panel / sandbox VMs
ether3	Access — IoT	v30	TV, Hue Bridge, smart plugs
ether4	Access — Guest AP	v40	Guest wireless AP (visitor SSID)
ether5	Spare / out-of-band	v10	Mgmt patch (optional)

Tailscale Tailnet, VPN Torrent & DNS

Tailscale Tailnet

OPNsense runs the **Tailscale subnet router + exit node**, advertising **10.10.0.0/16**. Native clients run on MHV, the Media Server, NAS, WGR, plus phones and laptops. ACLs use tag-based policy: **tag:admin** reaches MGMT v10, **tag:user** is restricted to LAN v20. MagicDNS aliases are mapped to ***.d2.lol** hostnames via AdGuard Home split-horizon so the same DNS name resolves to the LAN IP at home and the 100.x address remotely.

VPN Torrent Path (VLAN 60)

Gluetun runs as a privileged LXC on MHV with its own NIC pinned to VLAN 60 (**10.10.60.60**). qBittorrent and Prowlarr are bound exclusively to **tun0**. OPNsense enforces a policy route: any traffic sourced from **10.10.60.0/24** that misses the WireGuard tunnel is dropped — true kill-switch. Sonarr/Radarr live on v20 and pull completed media from the NAS via SMB, then trigger Plex library refresh. The torrent client itself never has a path to v20 servers.

DNS & d2.lol

Cloudflare hosts the public **d2.lol** zone. Public records resolve to the WAN IP (or a Cloudflare Tunnel for selected services). Internally, **AdGuard Home** on OPNsense runs split-horizon: ***.d2.lol** resolves to LAN IPs, with **plex.d2.lol**, **nas.d2.lol**, **mhv.d2.lol**, etc. wired to their VLAN 20 addresses. TLS is provided by Caddy (or HAProxy + acme.sh) using a Cloudflare DNS-01 wildcard cert for ***.d2.lol**.

Test / Sandbox VLAN (50)

VLAN 50 is the “break things” lab: routed through the hEX with strict OPNsense rules that allow internet egress but block all inbound to v10/v20. Use it for nested Proxmox, Terraform/Ansible runs, k3s experiments, and OS images you don’t fully trust. DHCP **10.10.50.50–.200**; static reservations live in the runbook.